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Chinese scientists have developed an AI 'prosecutor' that can press charges. <https://digit.in/jan22-01>



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Intel Core i9-12900K Desktop Processor

Hybrid architecture works wonders

Intel is back with another desktop processor launch and this time, it's the 12th Gen Intel Core "Alder Lake" family of desktop processors. These are based on their new hybrid microarchitecture which clubs traditional cores which are now called P-Cores (Performance Cores) with E-Cores (Efficiency Cores). Here's how they perform.

SPECIFICATIONS

Interestingly, the base frequency for the P-Cores is a little lower than what we saw with the 11th Gen Rocket Lake desktop processors. Not that it matters a lot, processors have various power levels and sleep states that allow for conservation of power and we could very well see the 12th Gen E-Cores consuming even less power than what the 11th Gen cores did.

The relationship between cores and threads has also changed. Only P-Cores sport multithreading, so if you're trying to figure out the number of logical CPUs or threads then you need to double the number of P-Cores and add that to the number of E-Cores. Which is why the Core i9-12900K (8P/8E Cores) has 24 threads and the Core i5-12600K (8P/8E Cores) has 16 threads.

Depending on which RAM type get for your PC, you will experience different performance levels. Needless to say, DDR5 trumps DDR4.

PERFORMANCE

In the single-threaded benchmark, the Intel Core i9-12900K scores 740 points showing tremendous gains over its predecessor, the Intel Core i9-11900K. The AMD Ryzen 9 5950X is nearly 100 points behind with a score of 644.

Even in the multithreaded run, the Intel Core i9-12900K goes straight to the top of the chart with a score of 10363 putting it ahead of the AMD



Ryzen 9 5950X with a gap of 267 points. Intel's 24 threads seem to do a better job than AMD's 32 threads. It would be interesting to see how well it performs with DDR4 rather than DDR5. Vice-versa, we'd like to see what AMD can do with DDR5.

Handbrake benefits from both, high core count as well as high IPC. We track the average number of frames processed per second. We see Intel take the lead yet again. 7-Zip usually favours AMD's Zen 3 processors over the Intel processors. Our scores from the Intel 10th Gen, 11th Gen and now 12th Gen all show lower performance than Intel processors. However, the gap between Intel and AMD has been lessened by a decent amount.

GAMING (DISCRETE GPU)

In our gaming benchmarks, we run several new titles on the NVIDIA RTX 2080 Ti. There are a host of different games based on a variety of current gen game engines using different graphics APIs such as DirectX11, 12, Vulkan and OpenGL. The scores indicate that Intel is still the best for gaming with the 12900K leading the pack.

GPU PERFORMANCE

While Intel did show great improve-

ment in the iGPU performance with 11th Gen, it seems that the faster memory and faster DMI bus has improved gaming on the iGPU by a decent amount. The Intel Core i9-12900K is powered by Intel UHD 770 graphics which seems to just about equal the performance of the AMD Ryzen 3 3200G. It still has a long way to go before catching up with AMD's GPUs since the Ryzen 7 5700G took things to a new level.

We see a 47 per cent improvement with the Intel Xe-LP over the older Intel Iris integrated graphics. Compared to the AMD Ryzen 3 3200G, the Xe-LP with 32 EUs would be an improvement but compared to the AMD Ryzen 5 3400G, the Xe-LP would still lag behind.

Running games on the iGPU also shows that we're seeing a similar 50 to 60 per cent improvement in average FPS. Unfortunately, current-gen AAA games aren't playable which is

normal for all iGPUs but older AAA titles such as GTA V are playable on low settings.

VERDICT

Intel certainly has a winner in their hands. The Intel 12th Gen Core "Alder Lake" Desktop processors outperform the Zen 3 based Ryzen 5000 processors in most of the benchmarks that we've tested the processors on. The hybrid architecture approach seems to work wonders and the ball is now in AMD's court. For the enthusiasts, Alder Lake with DDR5 would be the go to combination for the sheer performance that it offers.

—Mithun Mohandas

SPECIFICATIONS

CORES: 16 | THREADS: 24 | BASE CLOCK (P/E): (3.2 / 2.4) GHz | BOOST CLOCK (P/E): (5.1 / 3.9) GHz | PROCESS NODE: Intel 7 | L3 CACHE: 30 MB | L2 CACHE: 14 KB per core | MAX TURBO POWER: 241 W.

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